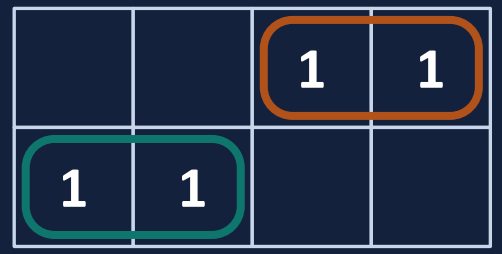
ENES 246 · DIGITAL LOGIC DESIGN

Lecture 6

Karnaugh Maps



$$F = x'y + xy'$$

Adjacency and Gray code · two-, three- and four-variable maps · prime implicants · product of sums · don't-cares

Reading: Mano & Ciletti, *Digital Design with an Introduction to the Verilog HDL*, 6th ed. — Chapter 3, §3.1–3.5

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Today’s roadmap

1 §3.1–3.2

Why a map

Adjacent cells differ in one variable

2 §3.3

Bigger maps

Group 1s in powers of two

3 §3.3

Prime implicants

Essential terms first, then cover the rest

4 §3.5

POS & don’t-cares

Group 0s; use X only when it helps

By the end of class you can…

- Label a map in Gray-code order and place minterms in it
- Form the largest legal groups, including wrap-around groups
- Read a product term from any group
- Identify essential prime implicants
- Find a minimum SOP and a minimum POS
- Use don’t-care conditions to simplify further

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Agenda for today’s session



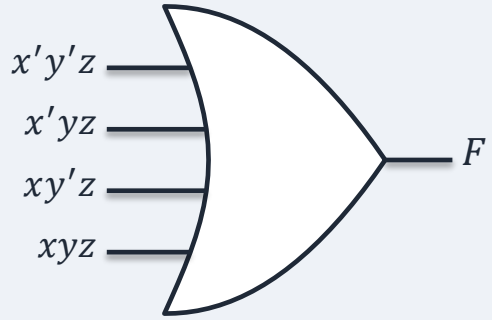
Elapsed time	Topic	In-class work
0:00 – 0:05	Welcome, roadmap, why minimize	—
0:05 – 0:15	Algebra has no stopping rule; the map idea	Discussion
0:15 – 0:40	Two- and three-variable maps, grouping rules	Examples 1, 2 · Your turn 1
0:40 – 0:50	Break	
0:50 – 1:15	Four-variable maps	Example 3 · Your turn 2
1:15 – 1:40	Prime implicants, product of sums, don’t-cares	Examples 4–6 · Your turn 3
1:40 – 1:50	Concept check and wrap-up	Homework preview

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Why minimize with a map?

§3.1–3.2

The same function, two circuits



Four 3-input ANDs + one 4-input OR (+ inverters)



…is just the wire z: no gates at all.

Algebra works, but…

$$F = x'y'z + x'yz + xy'z + xyz$$

$$= x'z(y' + y) + xz(y' + y) = x'z + xz = z$$

you choose which terms to pair, and nothing tells you when you have reached the minimum.

The map method

Draw the truth table as a grid in which neighbouring cells differ in exactly one variable.

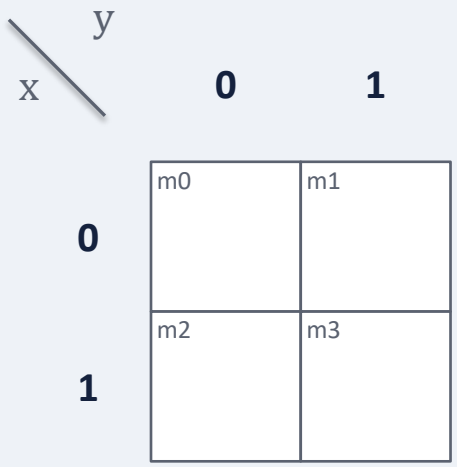
Then every pair of neighbouring 1s is one use of $xy + xy' = x$ — and you can SEE the largest groups.

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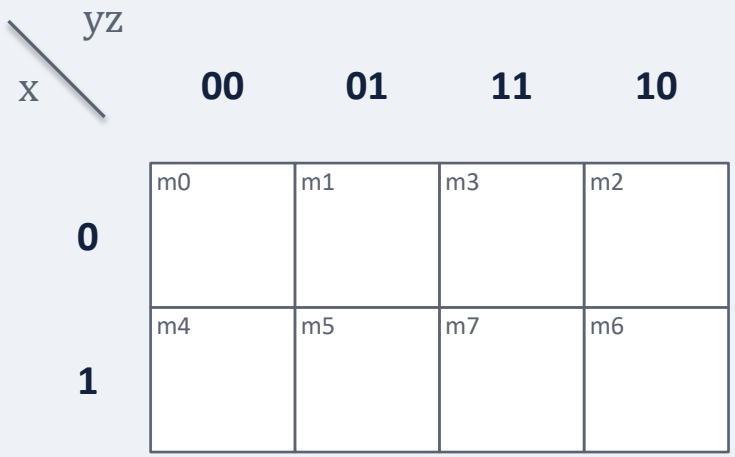
The map: adjacency by Gray code

§3.1–3.2

Two variables



Three variables



Gray-code order

Columns run 00, 01, 11, 10 — not 00, 01, 10, 11.

Each step changes one bit, including the wrap from 10 back to 00.

Adjacent means one variable changes

Cells that share an edge are adjacent — and so are the left and right edges of the map, which wrap around.

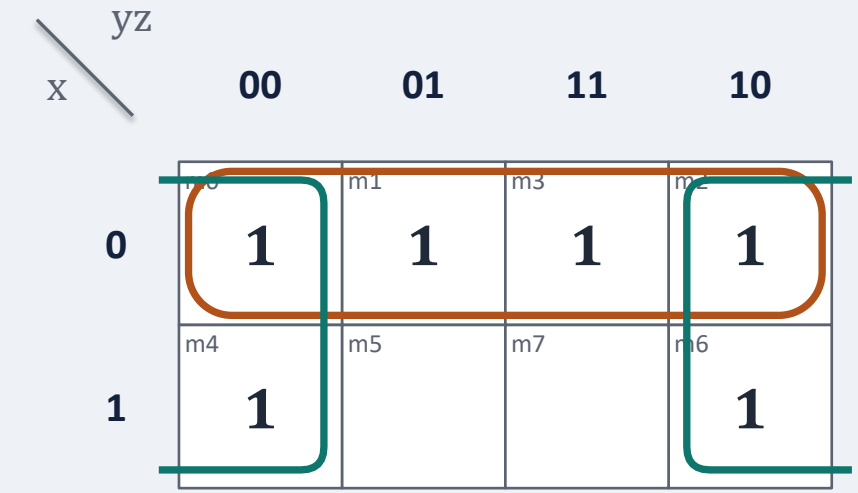
Note the order of the minterm numbers: m3 sits before m2 because its column is 11.

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Grouping rules and reading a group

§3.1–3.2

Legal groups



Copper: the whole top row, x = 0 → x'

Teal: the four outer cells (edges wrap), z = 0 → z'

$$F = x' + z'$$

The rules

1 Groups hold 1, 2, 4, 8 or 16 cells — a power of two.

2 Groups are rectangles; they may wrap around the edges.

3 Make every group as large as possible.

4 Use as few groups as possible; every 1 must be covered.

Covering a 1 twice is allowed.

Reading a group

A variable that is 1 in every cell of the group: write it (e.g. x).

A variable that is 0 in every cell: write its complement (x').

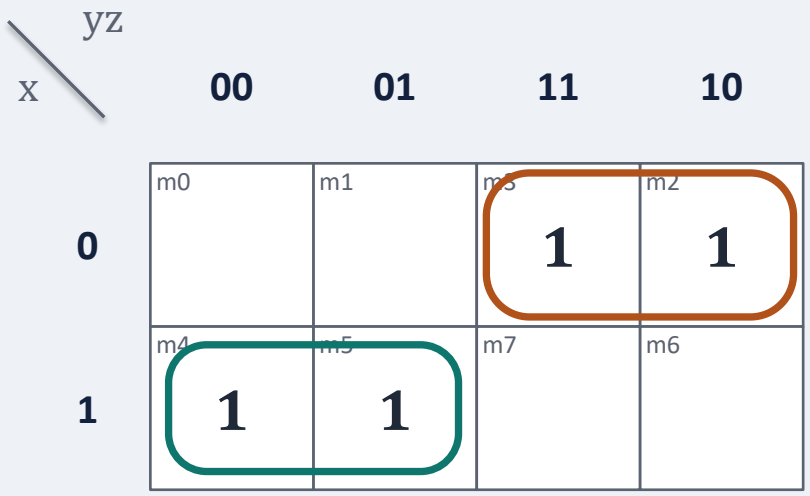
A variable that changes inside the group: drop it.

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Example 1 — A three-variable map

§3.1–3.2

EXAMPLE Simplify $F(x, y, z) = \Sigma m(2,3,4,5)$.



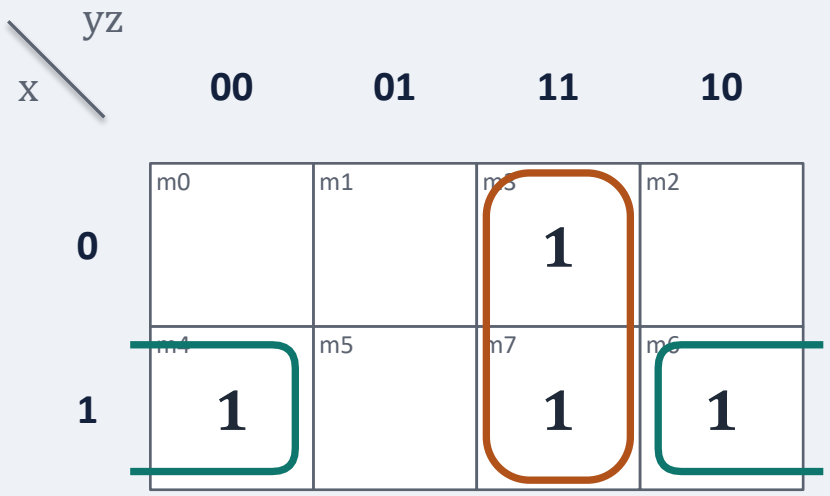
1. Place 1s in cells m2, m3, m4 and m5.
2. Look for a group of four: none (the pairs are in different columns).
3. Pair m2 (010) with m3 (011).
4. $x = 0, y = 1$ stay the same; z changes, so drop it $\rightarrow$ $x'y$
5. Pair m4 (100) with m5 (101): $x = 1, y = 0 \rightarrow xy'$
6. Every 1 is covered. OR the terms:
7. $F = x'y + xy'$ (four literals, and z does not appear)

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Example 2 — A wrap-around group

§3.1–3.2

EXAMPLE Simplify $F(x, y, z) = \Sigma m(3,4,6,7)$.



1. Place 1s in m3, m4, m6 and m7.
2. Look for a group of four: none forms a rectangle.
3. Pair m3 (011) with m7 (111): $y = 1, z = 1 \rightarrow yz$
4. m4 and m6 are in the outer columns: adjacent, since the map wraps.
5. m4 (100), m6 (110): $x = 1, z = 0$; y changes $\rightarrow xz'$
6. Algebra behind the wrap: $xy'z' + xyz' = xz'(y' + y) = xz'$
7. $$F = yz + xz'$$

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Your turn 1 — Three-variable maps

§3.1–3.2

YOUR TURN

≈ 8 min

x \ yz	00	01	11	10
0	m0	m1	m3	m2
1	m4	m5	m7	m6

Use this blank map (or draw your own) for each part.

Find the minimum sum of products.

(a) $F(x, y, z) = \Sigma m(0,2,4,5,6)$

(b) $F(x, y, z) = \Sigma m(0,1,3,7)$

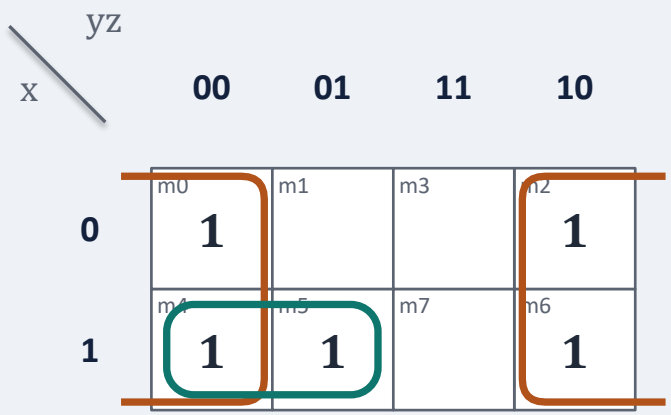
Hint: in (a), look for a group of four first.

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Your turn 1 — Solution

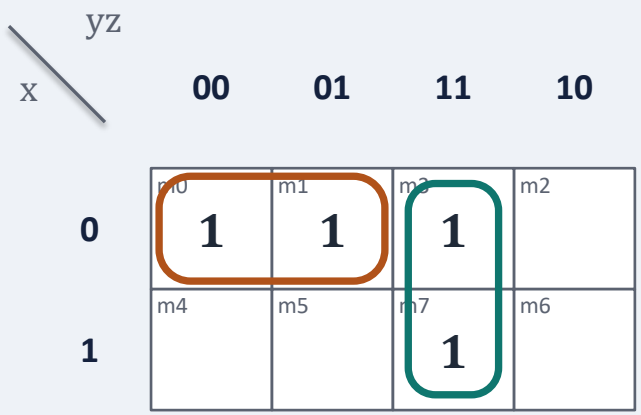
SOLUTION

(a)



1. Quad m0, m2, m4, m6 (outer columns, wraps)
2. Only $z = 0$ is constant → z'
3. m5 is left: pair m4 (100), m5 (101) → xy'
4. OR the terms: $F = z' + xy'$

(b)



1. No quad. m0 (000), m1 (001) → $x'y'$
2. m3 (011), m7 (111) → yz
3. m1, m3 could pair too, but both are covered.
4. OR the terms: $F = x'y' + yz$

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Break

10 minutes — we resume at the 0:50 mark

Next: four-variable maps

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The four-variable map

§3.3

wx \ yz	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10

Both directions use Gray code

Rows run 00, 01, 11, 10 — so the bottom row is m8…m10, not m12…m14.

The top and bottom edges wrap, as do left and right: the four corners are one legal group.

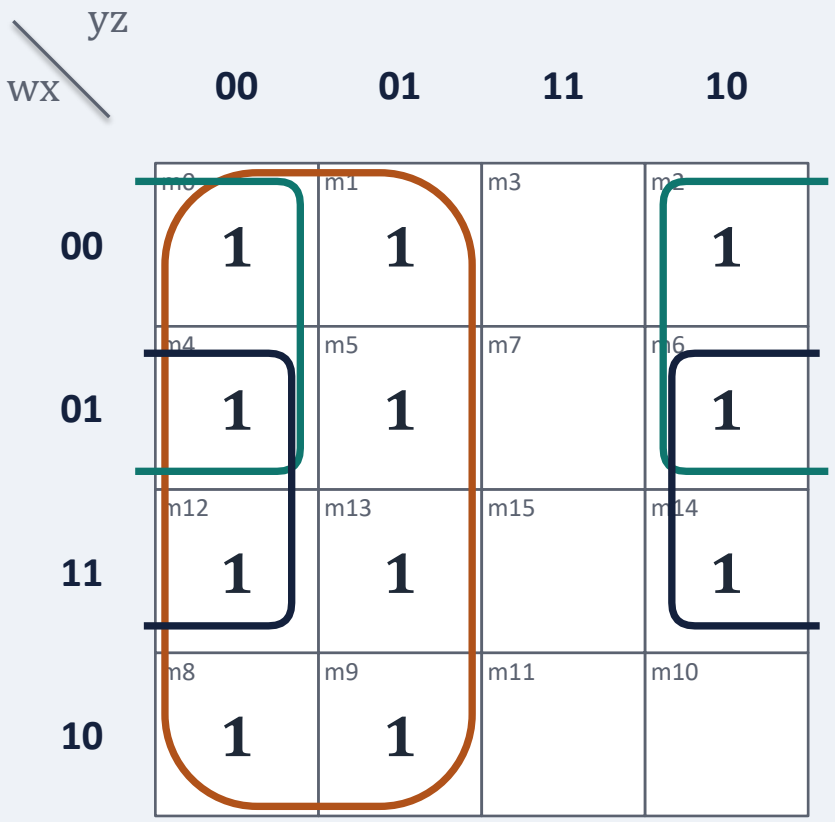
Group size	Variables eliminated	Literals left
1 cell	0	4
2 cells	1	3
4 cells	2	2
8 cells	3	1

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Example 3 — Eleven minterms

§3.3

EXAMPLE Simplify $F(w,x,y,z) = \Sigma m(0,1,2,4,5,6,8,9,12,13,14)$.



1. Place the eleven 1s.
2. Largest group first: the octet of the two left columns.
3. In the octet only $y = 0$ stays the same $\rightarrow$ y'
4. Left over: m2, m6 and m14, all in column 10.
5. Quad m0, m2, m4, m6 (rows 00–01, outer columns): $w = 0, z = 0$ $\rightarrow$ $w'z'$
6. Quad m4, m6, m12, m14 (rows 01–11, outer columns): $x = 1, z = 0$ $\rightarrow$ xz'
7. $F = y' + w'z' + xz'$ (11 minterms → 3 terms, 5 literals)

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Your turn 2 — Expression to map

§3.3

YOUR TURN

≈ 10 min

AB \ CD	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12	m13	m15	m14
10	m8	m9	m11	m10

Simplify

$$F = A'B'C' + B'CD' + A'BCD' + AB'C'$$

(a) Expand each term into minterms and place them on the map.

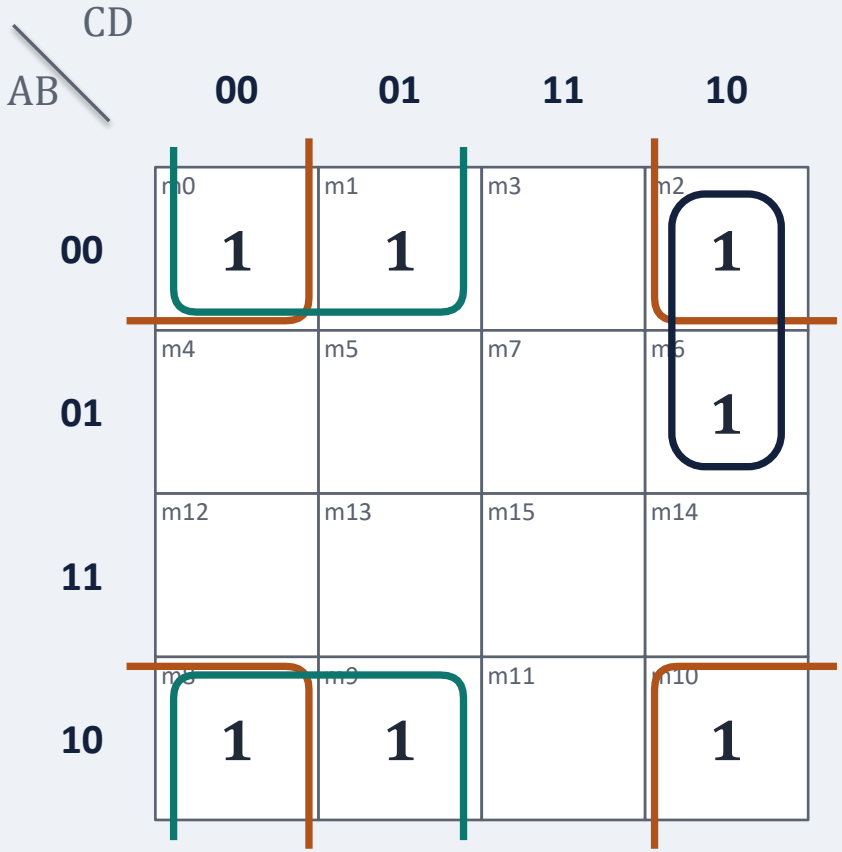
(b) Find the minimum sum of products.

Hint: a term with three literals covers two cells.

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Your turn 2 — Solution

SOLUTION



1. (a) $A'B'C'$ lacks D: $A'B'C'D' + A'B'C'D$ = m0, m1
2. $B'CD'$ lacks A: $A'B'CD' + AB'CD'$ = m2, m10
3. $A'BCD'$ is already a minterm: m6. $AB'C' \rightarrow$ m8, m9
4. So $F = \Sigma m(0,1,2,6,8,9,10)$
5. (b) Four corners m0, m2, m8, m10: $B = 0, D = 0 \rightarrow B'D'$
6. m0, m1, m8, m9: $B = 0, C = 0 \rightarrow B'C'$
7. m6 is left: pair with m2 (0010, 0110) $\rightarrow A'CD'$
8. $F = B'D' + B'C' + A'CD'$ (14 literals → 7)

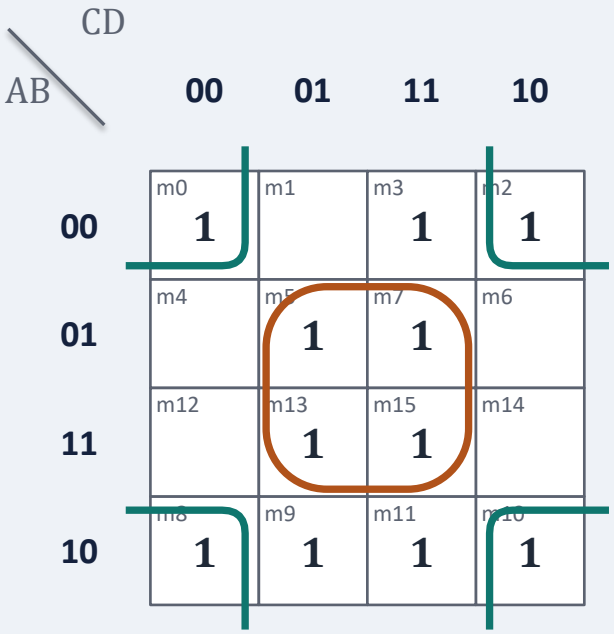
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Example 4 — Essential prime implicants

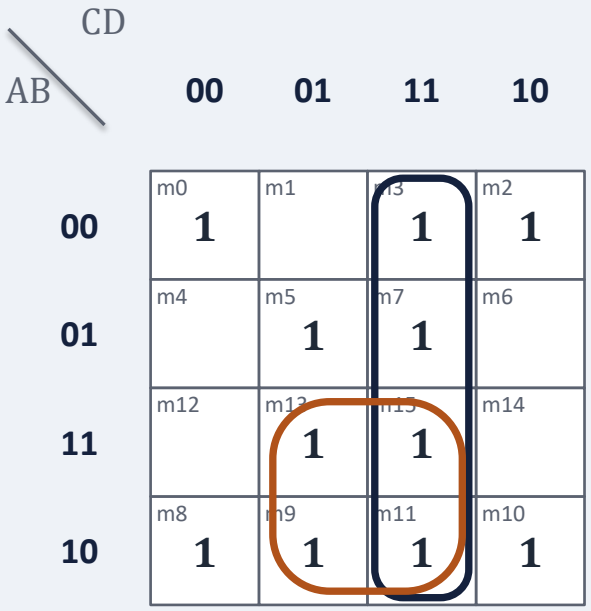
§3.3

EXAMPLE Simplify $F(A,B,C,D) = \Sigma m(0,2,3,5,7,8,9,10,11,13,15)$.

Essential prime implicants



Covering the rest



1. **Prime implicant**: a group that cannot be made larger.
2. Find 1s that only ONE prime covers.
3. m5: only BD covers it → BD is **essential**
4. m0: only the corners $B'D'$ → **essential**
5. Still uncovered: m3, m9, m11.
6. m3: pick CD (or $B'C$).
7. m9, m11: pick AD (or AB').
8. $F = BD + B'D' + CD + AD$ — one of four minimum answers

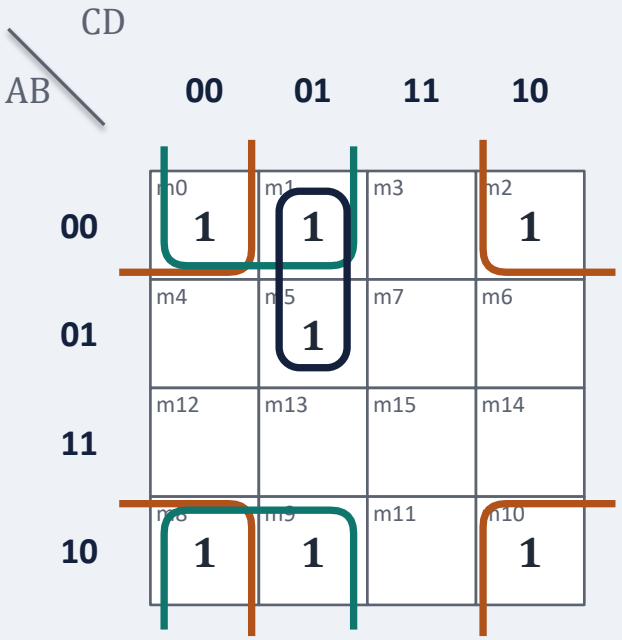
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Example 5 — Product of sums

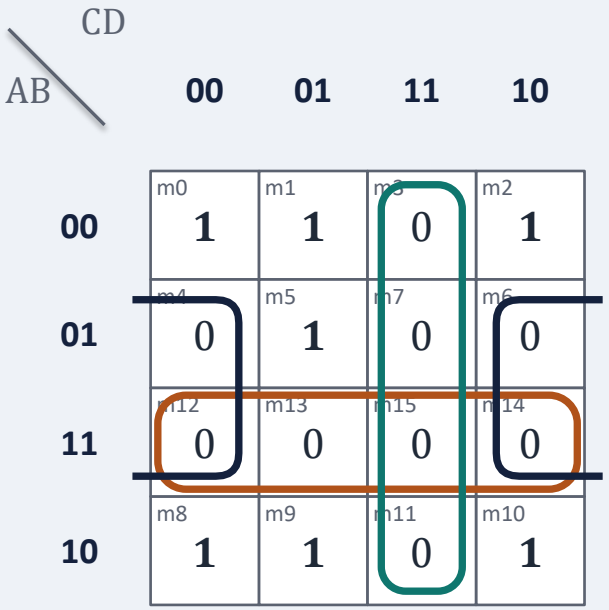
§3.4

EXAMPLE Find the minimum SOP and POS of $F(A,B,C,D) = \Sigma m(0,1,2,5,8,9,10)$.

Group the 1s (SOP)



Group the 0s (POS)



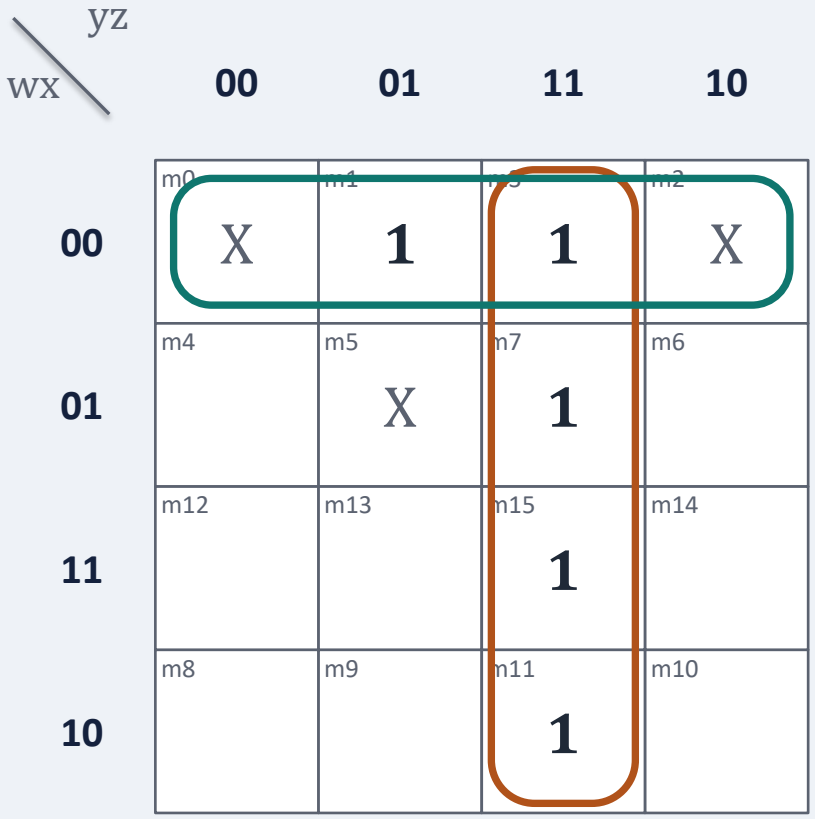
1. SOP: group the 1s: $B'D'$, $B'C'$, $A'C'D$
2. $$F = B'D' + B'C' + A'C'D$$
3. POS: write 0s in the empty cells.
4. Group the 0s: AB, CD, BD'
5. That is $F' = AB + CD + BD'$
6. DeMorgan: $F = (AB)'(CD)'(BD')'$
7. $$F = (A' + B')(C' + D')(B' + D)$$
8. Check m5 (0101): both forms give 1 ✓

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Example 6 — Don’t-care conditions

§3.5

EXAMPLE Simplify $F(w,x,y,z) = \Sigma m(1,3,7,11,15)$ with $d(w,x,y,z) = \Sigma m(0,2,5)$.



1. A don't-care (X) never occurs: we may treat it as 0 or 1.
2. Place 1s in m1, m3, m7, m11, m15 and X in m0, m2, m5.
3. Column $yz = 11$: m3, m7, m11, m15: $y = 1, z = 1 \rightarrow yz$
4. m1 is left. Treat the X at m0 and m2 as 1.
5. The top row is now a quad: $w = 0, x = 0 \rightarrow w'x'$
6. m5 is not needed: leave that X as 0.
7. $$F = yz + w'x'$$

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Your turn 3 — A BCD function

§3.5

YOUR TURN

≈ 8 min

AB \ CD	00	01	11	10
00	m0	m1	m3	m2
01	m4	m5	m7	m6
11	m12 X	m13 X	m15 X	m14 X
10	m8	m9	m11 X	m10 X

The input $ABCD$ is a BCD digit, so the codes 10–15 never occur.

$$F = \Sigma m(0,2,4,5,6) + d(10,11,12,13,14,15)$$

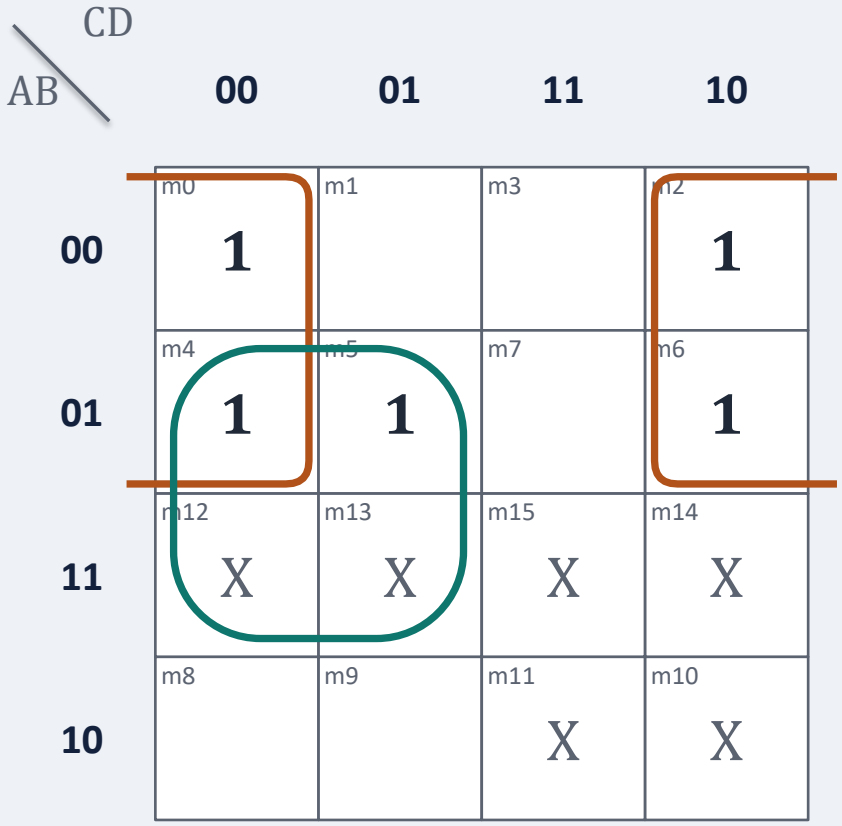
Find the minimum sum of products.

Watch out: m8 is not a don’t-care.

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Your turn 3 — Solution

SOLUTION



1. Place 1s in m0, m2, m4, m5, m6 and X in m10–m15; m8 is 0.
2. Quad m0, m2, m4, m6: $A = 0, D = 0 \rightarrow A'D'$
3. m5 is left. Treat the X at m12 and m13 as 1.
4. Quad m4, m5, m12, m13: $B = 1, C = 0 \rightarrow BC'$
5. Trap: D' alone would also cover m8, which must be 0.
6. $$F = A'D' + BC'$$

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Concept check

1. Why do the columns of a map run 00, 01, 11, 10?

2. Can a group contain 6 cells? Can it contain the four corners?

3. What makes a prime implicant essential?

4. How do you get a POS from the map, and when is it cheaper?

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Key takeaways

Gray-code order

00 01 11 10

Group sizes

1, 2, 4, 8, 16

A group of 2^k cells

drops k variables

Essential PI

only cover of some 1

POS

group 0s → F' → DeMorgan

Don't-care X

use only if it helps

The map gives a guaranteed minimum for up to four variables.

Homework #3 due at the start of Lecture 7

This afternoon's lab: Lab B — Verilog and the Nexys 4 DDR: switch to LED

Next: NAND and NOR implementations and the XOR function

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